# CONSTRUCT CONSISTENCY OF IN-DECODING CONFIDENCE READOUTS FOR RISK-GATED MASKED-DIFFUSION DECODING

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## ABSTRACT

Masked-diffusion decoders expose several token-confidence summaries, but choosing among them need not produce the same decode ranking or selective-prediction utility. On one frozen LLaDA-8B-Instruct $\times$ GSM8K bed ($974$ distinct problems, one decode each), we compare last-revision (`commit`), first-exposure (`first`), and within-block mean (`all`) readouts using decode-clustered inference. In a post-alignment exploratory contrast with paired bootstrap inference, AUROC(`commit`)$-$AUROC(`all`) is $+0.0278$ in rank units. By contrast, every paired risk difference on the coverage grid $\{0.2, 0.5, 0.8\}$ is small and has a zero-crossing CI within the intra estimand (Table 5). The early/late equality at coverage $0.2$ is a shared $1/195$ lattice observation, not evidence of identical retained sets, and a noise-model-sensitive reliability bound cannot robustly separate $\rho = 1$ from $\rho < 1$. Thus this frozen bed separates `commit` and `all` in rank units, but it establishes neither practical equivalence nor a generally null selection increment. The paper instead provides an auditable frozen-stream analysis and specifies the replication needed to test this bounded result.

## 1 INTRODUCTION

Selective prediction trades coverage for lower conditional risk by abstaining on low-confidence examples (Chow, 1970; El-Yaniv & Wiener, 2010; Bartlett & Wegkamp, 2008; Geifman & El-Yaniv, 2017). A masked-diffusion language model revises positions in parallel, so a decode admits several plausible confidence summaries: last-revision confidence (`commit`), first-exposure confidence (`first`), and the within-block mean (`all`). Their ranking quality need not imply an observable difference in risk at a chosen coverage, and non-significance cannot by itself establish absence (Altman & Bland, 1995).

We study that measurement problem on a single frozen event stream, not model internals and not best-of-$k$ generation. Each of the $974$ prompts has exactly one decode, making prompt, problem, and decode the same independent cluster. The central analysis aligns the treatment contrast on both readouts: `commit`$-$`all` AUROC and risk@coverage use the same decodes and paired resamples. A second, explicitly different AUROC-over-random estimand is retained only to connect the result to the registered discrimination analysis.

Our contributions are threefold: (i) an aligned, same-arm comparison that places an exploratory rank-separation estimate beside paired risk increments; (ii) a calibrated joint-endpoint analysis showing that the three-anchor family design alone does not explain the unresolved risk increment; and (iii) boundary audits for lattice equality, retained-set overlap, residual-covariance sensitivity, practical equivalence, and experiment recoverability. Every claim remains restricted to this event stream and the recorded estimator.

Figure 1 previews the measurement flow and marks the two non-interchangeable contrasts used below.

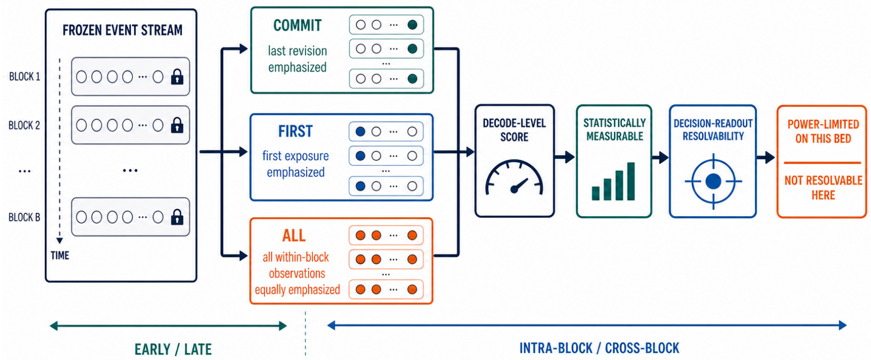


Figure 1: Measurement flow from one frozen event stream to three confidence summaries and a decode-level selection score. **Takeaway:** the `commit`-vs-`all` incremental selection utility is not resolvable at this bed's power, a distinct estimand from the registered AUROC-vs-random discrimination lift. Conceptual figure; it contains no experimental quantities beyond the captioned, evidence-linked takeaway.

Table 1: Estimand crosswalk. H2 is a conservative robustness flag, not the nominal detection test. **Takeaway:** units, contrasts, and gates are never interchanged.

| Estimand | Contrast and unit | Scale | MDE tag | Applicable gate |
|---|---|---|---|---|
| Registered rank | `commit` vs random ordering | AUROC | $\mathrm{MDE}_{80}^{A}$ | CI/p; H2 flag |
| Aligned rank | `commit` vs `all`, paired decode | AUROC | $\mathrm{MDE}_{80}^{A}$ | CI/p; H2 flag |
| R3 utility | `commit` vs `all`, paired decode | risk | $\mathrm{MDE}_{80}^{R,c}$ | CI/p; TOST; H2 |
| R1/R4/R5 | early vs late window, paired decode | risk | $\mathrm{MDE}_{80}^{EL,c}$ | Holm(3); TOST |

## 2 Measurement and Inference

### 2.1 Readouts and Estimands

For window $W \in \{E, L\}$ (blocks 0–3 or 4–7), readout $r$, and coverage $c$, $\mathrm{risk}_{W,r}(c)$ is the error among the top $k = \max(1, \mathrm{round}(974c))$ decodes ranked by the window-mean readout. The early/late quantity used by R1, R4, and R5 is $\delta_{\mathrm{EL},r}(c) = \mathrm{risk}_{E,r}(c) - \mathrm{risk}_{L,r}(c)$. R3 instead uses the paired, within-block arm difference

$$\Delta^{\mathrm{intra}}(c) = \mathrm{risk}^{\mathrm{intra}}_{\texttt{commit}}(c) - \mathrm{risk}^{\mathrm{intra}}_{\texttt{all}}(c), \tag{1}$$

where risk is computed per block and averaged over eight blocks.

We stress that the registered two halves use different contrasts and are never merged: the AUROC half compares `commit` with a *random-ordering baseline* (ranking units), whereas the risk@coverage half compares `commit` with `all` (selection-utility units). A measurable AUROC-over-random lift therefore does not imply a resolvable `commit`-vs-`all` selection gain; these are distinct estimands on different units, **distinct contrasts, not a dissociation**. Table 1 is the sole estimand crosswalk used by subsequent forward references. The exact aggregation and inference procedures—including tie, revision, and block-averaging rules and a hand-verifiable worked example—appear in Appendix A.

The single-readout window-restricted benefit $\mathrm{gain}_W = \mathrm{base} - \mathrm{risk}_{W,r}$ is about $+8.7$ percentage points at $c = 0.2$. It is distinct from the A0 full-window rig control $\mathrm{gain}_{\mathrm{full}} = +0.2126$ at the same coverage (Table 12). It shows that each readout can risk-gate relative to no selection; it is not the `commit`-vs-`all` increment.

### 2.2 Gate and power conventions

Inference resamples decodes as clusters and preserves the readout and coverage vector within a resample (Field & Welsh, 2007). Registered anchor families use Holm's sequential procedure over $c \in \{0.2, 0.5, 0.8\}$ (Holm, 1979). A conventional two-sided 95% CI and $p_{\text{two-sided}}$ report detection. Practical absence is evaluated separately by two one-sided tests (TOST) at the registered practical bound $\delta^* = 1$ percentage point (E0) (Schuirmann, 1987).

The historical H2 flag uses $\mathrm{MDE}_{80} = 2.8016\,\widehat{\mathrm{SE}}$ and labels $\mathrm{MDE}_{80}/|\hat{\Delta}| \geq 1$ as `PL`. Algebraically this is a two-sided $z$ threshold with de facto $\alpha = 2\{1 - \Phi(2.8016)\} \approx 0.0051$, stricter than nominal $0.05$. We retain H2 only as a conservative 80%-power robustness criterion. Observed-SE MDE80 is a descriptive sensitivity/design reference, never a post-hoc substitute for the CI/p detection decision (Hoenig & Heisey, 2001). Thus the cross-block row that excludes zero at nominal 95% is *detected but below the robustness threshold*, not a non-detection. Only TOST can support a practical-equivalence statement.

**Statistical conventions and prior art.** Holm correction, clustered resampling, TOST, and the distinction between absence of evidence and evidence of absence are established guidance (Holm, 1979; Field & Welsh, 2007; Schuirmann, 1987; Altman & Bland, 1995). Our H1–H6 appendix restates those principles for readout-to-selection claims. The practice combination specific to this study is to co-report the full-width, scale-tagged H2 sensitivity and to re-run the entire CI-plus-family procedure under label permutation (Table 7); neither component is claimed as a new statistical test.

Table 2: Notation glossary. R1–R5 are the five locked claims audited in Table 3; scopes and verdict labels are fixed before inference and may not be expanded post-hoc. **Takeaway:** a claim referenced in the prose below always resolves to one of these rows.

| Symbol | Plain-language definition |
|---|---|
| R1 | Equal lattice risks shared by two readouts at any fixed coverage; says nothing about set identity |
| R2 | Whether residual-covariance ratio $\rho = 1$ (latent model collapses) robustly separates from $\rho < 1$ |
| R3 | Aligned rank separation of `commit` vs `all` readouts and their selection-increment |
| R4 | Desaturation (intra-decode confidence pull) and coverage-localized risk differences after alignment |
| R5 | High-selectivity early-vs-late risk pattern inside a Holm-corrected anchor family |
| PL | $\mathtt{MDE}_{80}/|\hat{\Delta}| \geq 1$: observed SE too coarse to detect the measured effect at 80% power |
| RES | Residual covariance; the R2 construction that compares raw to covariance-corrected risks |
| H2 | Historical strict flag: reports $\mathrm{MDE}_{80} = 2.8016\,\widehat{\mathrm{SE}}$; robustness only, not a decision criterion |
| E0 | Registered 1-pp practical-equivalence bound used by TOST |
| $\mathrm{MDE}_{80}$ | Minimum detectable effect at 80% power under the observed SE |

## 3 Results

Table 3 is the claim-level map for the detailed result sections.

### 3.1 R1 and R2: lattice and latent-model boundaries

**R1.** The lowest-coverage equality is one shared lattice observation, not three independent working instances. The tie and retained-set audits in Table 4 show that identical lattice risk does not imply an identical retained set; the higher anchors also exclude wholesale pipeline collapse.

**R2.** Under the assumed residual-covariance correction, the point result lies below unity, but the sensitivity rows in Table 4 show that a modest covariance perturbation erases the rejection. We therefore cannot robustly separate $\rho = 1$ from $\rho < 1$. A 61-point residual-covariance sweep over the frozen bed has now been executed in this refine pass and is reported in full in Appendix D; it is an *analysis*-level sensitivity statement computed over the frozen asset, not a causal or model-internal claim.

Table 3: R1–R5 verdict matrix. **Takeaway:** the aligned rank separation is an exploratory, post-alignment estimate; risk utility and boundary claims remain bounded.

| Key | Claim | Gate | Detail |
|---|---|---|---|
| R1 | Equal lattice risks; retained sets differ | boundary audit | §3.1 |
| R2 | $\rho = 1$ vs $\rho < 1$ not robustly separated | sensitivity-open | §3.1 |
| R3 | Rank separated (exploratory, post-alignment); risk increment unresolved | CI/TOST/H2 | Table 5 |
| R4 | Desaturation changes variance; readout-level risk differences remain coverage-dependent | paired-diff bootstrap | §3.3 |
| R5 | High-selectivity pattern lacks family support | Holm(3) empty | Figure 2 |

Table 4: R1/R2 boundary audit, with comparable values transferred from the frozen tie, retained-set, and residual-covariance assets. **Takeaway:** equal lattice risk does not establish identical selection, and the latent-correlation result remains sensitivity-open.

| Key | Audit | Frozen values | Reading |
|---|---|---|---|
| R1 | Lattice | $c = .2$: all readouts $\delta_{\mathrm{EL}} = -15/195 = -.076923$ | One shared observation |
| R1 | Tie stability | tie mass $\leq .0051$; 200 breaks: Jaccard 1.0, risk SD 0 | Stable to recorded tie breaks |
| R1 | Retained sets | tie mass 0; Jaccards $.569/.690/.749$ (commit–first/commit–all/first–all) | Equal risk, different sets |
| R1 | Other anchors | $c = .5/.8$: the three point estimates diverge | No wholesale collapse |
| R2 | Corrected point | upper bound .979; reliabilities $.755/.678$ (commit/all) | Below unity under the assumed correction |
| R2 | Sensitivity | 6.5–10% over-estimate erases rejection; point breakeven $m \approx .905$; CI-upper returns to 1 at $m \approx .94$; 61-pt sweep over $[.85, 1.15]$ now run on frozen bed (r211) | No ground-truth $\rho$; analysis-level statement |

### 3.2 R3: ALIGNED RANK AND SELECTION CONTRASTS

**R4 is a lattice observation of R1, not independent evidence.** R1 reports a $c = 0.2$ lattice value shared by all readouts ($\delta_{\mathrm{EL}} = -15/195 = -0.076923$, Table 4). R4 re-reads that same frozen cell through the paired-difference resample of §3.3; the $-15/195$ underlying observation is identical. R4 therefore localizes *where* a shared lattice point sits inside its resampled distribution; it does not supply a second observation supporting R1. When support counts are tallied elsewhere in this paper R4 is explicitly excluded as an *independent* anchor of R1.

Table 6 keeps the registered AUROC-to-risk bridge separate from the aligned paired contrast in Table 5. The bridge is a scale sensitivity, not a cross-estimand equality; its assumptions are listed in Appendix C. The joint endpoint and resample diagnostics likewise do not flip R3, while the cross-block row remains nominally detected but below the separate H2 robustness threshold.

Bootstrap centering is not driving the family decision: observed-centered, null-centered, and permutation-calibrated twin-$p$ variants all yield an empty Holm(3) survivor set (`decision_flip=False`). Here "twin $p$" denotes the bootstrap tail calculation defined in Appendix B, not a second hypothesis family.

Table 7 records the calibration check, and Table 8 separates the equivalence result from the next-bed replication targets.

### 3.3 R4 AND R5: DESATURATION AND COVERAGE LOCALIZATION

**R4.** Table 9 reports the direct paired-difference bootstrap between the `first` and `commit` readouts at each coverage, replacing any indirect CI-overlap argument; because both readouts derive from the same decodes, the paired difference is the appropriate test. The $c = 0.2$ lattice equality is the same shared observation as R1, not independent support. Saturation compresses variance, while the readout-level risk difference remains coverage-dependent: not established at the selective anchors, and a small positive shift at the intermediate one.

**R5.** The earlier mixed-stratum ratio is replaced by the like-for-like coverage estimates grouped in Table 9, shown across the full profile in Figure 2, and compared with the registered anchor grid in

Table 5: Headline aligned contrast from the frozen r204 paired-contrast asset (exploratory, post-alignment analysis). Two panels share the same frozen, block-mean decode scores and a decode-cluster bootstrap ($B = 2000$; r217 recomputation `results/r217_readout_auroc_aurc.json`). The upper panel reports absolute readout-level AUROC and AURC (risk–coverage area over $c \in [0.05, 1]$, trapezoid, mean risk); the lower panel reports the paired `commit−all` contrasts, whose AUROC reference uses the normal SE implied by the reported percentile-CI width and whose risk ratios use the shared r205 paired bootstrap. **Takeaway:** within one estimand, contrast, and paired resampling scheme, the two readouts separate in rank units while all three risk increments have zero-crossing CIs; the separation is reported as exploratory, not confirmatory.

| Quantity | Point | 95% CI | Note |
|---|---|---|---|
| *Absolute readout level (same frozen scores)* | | | |
| Absolute AUROC `commit` | 0.7377 | $[0.7053, 0.7696]$ | |
| Absolute AUROC `all` | 0.7100 | $[0.6734, 0.7443]$ | |
| Absolute AUROC `first` | 0.6970 | $[0.6592, 0.7312]$ | |
| Absolute AURC `commit` | 0.1603 | $[0.1369, 0.1884]$ | |
| Absolute AURC `all` | 0.1797 | $[0.1537, 0.2100]$ | |
| Absolute AURC `first` | 0.1895 | $[0.1621, 0.2207]$ | |
| *Paired `commit−all` contrast* | | | |
| Paired AUROC $\Delta$ | $+0.0278$ | $[0.0123, 0.0438]$ | H2 MDE80$/\lvert\Delta\rvert = 0.81$; `RES` |
| Paired risk @ 0.2 | $-0.0058$ | $[-0.0199, 0.0058]$ | H2 2.79; `PL`; CI $\ni 0$ |
| Paired risk @ 0.5 | $-0.0067$ | $[-0.0131, 0.0008]$ | H2 1.62; `PL`; CI $\ni 0$ |
| Paired risk @ 0.8 | $-0.0008$ | $[-0.0056, 0.0022]$ | H2 3.60; `PL`; CI $\ni 0$ |

Table 6: R3 scale-bridge and joint-endpoint diagnostics. Values sharing a row belong to one diagnostic; aligned AUROC and risk increments remain in Table 5. **Takeaway:** neither the descriptive bridge nor the pooled diagnostics resolves the aligned selection increment.

| Diagnostic | Frozen values | Reading |
|---|---|---|
| Registered bridge | AUROC lift $+.0144$; AUROC MDE80 .0069; projected risk $\approx .012$ vs risk MDE80 .017 ($\approx .71\times$) | Descriptive scale sensitivity; not aligned $+.0278$ |
| Joint primary | point $-.004415$; 95% CI $[-.009971, .000521]$; SE .002662; $z = -1.7836$; $p = .07449$ | Zero-crossing pooled endpoint |
| Resample signs | all-three-negative fraction .6236; joint sign-flip $p = .7246$; Wilcoxon $p = .08826$ | No decision flip |
| Cross-block | $z = 2.24$; $p \approx .025$ | Nominally detected; H2 robustness not passed |

Table 13. The descriptive H2 flag is unchanged, but neither view establishes practical equivalence: no R5-specific TOST accepts E0, and the registered Holm(3) family remains empty.

## 4 Experimental Setup and Reproducibility

The frozen bed is LLaDA-8B-Instruct (Nie et al., 2025) on GSM8K (Cobbe et al., 2021). Table 10 groups the complete frozen schedule, grading construct, and answer-extractor audit. The frozen artifacts do not retain the *original extraction-stage* labels, so the original path each answer took cannot be verified; a post-hoc rerun of the canonical extractor on the frozen prediction text (in this refine pass) populated the fallback branch census, the unparseable count, and an $n = 200$ manual-equivalence audit. That rerun supplies the reported numbers; it does not recover the original label provenance, so the manual audit counts as a noise *upper bound*, not a calibrated gold estimate.

**First-class recoverability limitation.** The remasking policy defines the `commit`/`first` distinction and is therefore an experimental decoder hyperparameter, not incidental metadata. It was not recorded; neither were the generation seed or prompt template. Consequently nobody, including the authors, can re-decode this bed. Every conclusion is conditional on the frozen event stream as given.

The analysis layer is deterministic. The event stream is identified by SHA256 `6e39e0ad3d4a42340ab6ef31474a20a2bdf7356dac9c9dbf6c0de63968b85972`;

Table 7: Decision stability of the R3 risk anchors under label permutation (1000 label permutations, inner $B = 1000$ paired bootstrap). Achieved per-anchor false-positive rates are $0/1000$ at every coverage, whose exact one-sided Clopper–Pearson upper bound is $\approx 0.003$. The permutation procedure tests a label-independence null, which is not the same as an equal-performance null hypothesis; we therefore report the outcome as *decision stability under this permutation procedure*, not as calibration of the detection test against the equal-performance alternative. The null-calibrated central band does not create a PL→RES flip. **Takeaway:** all three R3 decisions are unchanged by this permutation stress test.

| $c$ | Achieved FP (CP upper) | Calibrated 95% band | Excludes 0 | Flip |
|---|---|---|---|---|
| $0.2$ | $0/1000\ (< 0.003)$ | $[-0.0319, 0.0190]$ | no | no |
| $0.5$ | $0/1000\ (< 0.003)$ | $[-0.0199, 0.0076]$ | no | no |
| $0.8$ | $0/1000\ (< 0.003)$ | $[-0.0097, 0.0065]$ | no | no |

Table 8: Equivalence and next-bed design at the R3 decision anchor. E0 is the practical bound $\delta^* = 1$ pp; it was fixed after the original effect estimates were observed, before the TOST recomputation, and is retained as the target for subsequent prospective replication. **Takeaway:** lower-coverage absence is not established; the next bed needs roughly two to three times as many decodes.

| Check | Anchor | Result | Decision |
|---|---|---|---|
| TOST | $c = 0.2$ | 90% CI $[-0.018, 0.004]$ | Inconclusive at E0 |
| TOST | $c = 0.5$ | 90% CI $[-0.012, -0.001]$ | Inconclusive at E0 |
| TOST | $c = 0.8$ | 90% CI $[-0.005, 0.002]$ | Equivalent within $\pm 1$ pp |
| Design | MDE80 $= 1.2$ pp | $n \approx 1960$ | Replication target |
| Design | MDE80 $= 1.0$ pp | $n \approx 2820$ | Replication target |

its metadata table by `2b5c1f9d116257a7ccab6403c619cf0b74526472e4cabe11 0ed8a97cdf6258a5`. Scripts, seeds, schema, and a co-located manifest define the analysis release; release of the event-stream bundle remains subject to the project data steward's decision.

**Ethics.** This frozen benchmark analysis involves no human subjects, private user data, or live traffic. Deployment would require recalibration on the target distribution, coverage range, and cost function.

## 5 Related Work

Three literature streams leave the present question open. Discrete and masked diffusion work establishes denoising in discrete state spaces and competitive language modeling (Austin et al., 2021; Sahoo et al., 2024; Nie et al., 2025; Han et al., 2023; Ye et al., 2025); it does not establish which internal confidence aggregation is a stable decode-level selective score. Failure prediction, calibration, and LLM self-assessment show that maximum probability, verbalized confidence, or self-evaluation can track correctness (Hendrycks & Gimpel, 2017; Guo et al., 2017; Lin et al., 2022; Kadavath et al., 2022; Kuhn et al., 2023); they leave window and aggregation choice largely implicit. Selective classification and risk–coverage evaluation formalize the utility target (Chow, 1970; El-Yaniv & Wiener, 2010; Geifman & El-Yaniv, 2017; 2019; Ding et al., 2020), but not the measurement provenance of an iterative decoder.

**Closest confidence-gated decoding work.** Fast-dLLM uses token confidence as an actionable per-token signal for threshold-gated parallel unmasking (Wu et al., 2025). Our outcome is decode-level risk after aggregating that signal. A per-token action can therefore be useful while a difference between two decode-level summaries is unresolved. Early halting and self-consistency address stopping or multiple-sample agreement (Lo Cicero Vaina et al., 2023; Wang et al., 2023); the present $k = 1$ bed instantiates neither a candidate pool nor repeated generation. The LLaDA line itself couples per-token confidence with the remasking schedule (Nie et al., 2025; low-confidence remasking and its confidence-aware variants), but does not treat any in-decoding aggregation as a decode-level selective score against a paired, clustered risk endpoint; that is precisely the junction measured here.

Table 9: R4/R5 desaturation and localization audit. R4 rows report the direct paired-difference bootstrap of risk(`first`)−risk(`commit`) on identical decode-cluster resamples ($B = 2000$), with early-window rows shown in parentheses. **Takeaway:** desaturation changes variance, the readout-level risk difference is coverage-dependent rather than uniformly absent, and the high-selectivity pattern remains power-limited without a family survivor.

| Key | Audit | Values | Reading |
|---|---|---|---|
| R4 | Paired diff $c = .2$ | $+.0410$ [0, .0923] (early: $+.0103$ [$-.0256$, .0615]) | CI touches zero; R1 lattice shared |
| R4 | Paired diff $c = .5$ | $+.0452$ [.0164, .0678] (early: $+.0411$ [.0164, .0678]) | `first` keeps more errors |
| R4 | Paired diff $c = .8$ | $-.0013$ [$-.0128$, .0141] (early: 0 [$-.0141$, .0141]) | Not established |
| R4 | Late-window spread | SD `first` $\approx$ .20 vs `commit` .055 | Saturation compresses variance |
| R5 | Like-for-like $c = .6$ | $\delta_{\rm EL} = -.012414$; SE .008983; MDE80/$\lvert\delta\rvert = 2.027$ | H2 power-limited |
| R5 | Like-for-like $c = .8$ | $\delta_{\rm EL} = -.002246$; SE .005683; MDE80/$\lvert\delta\rvert = 7.087$ | H2 power-limited |
| R5 | Family/equivalence | Holm(3): no survivor; no R5-specific TOST accepts E0 | No practical-equivalence claim |

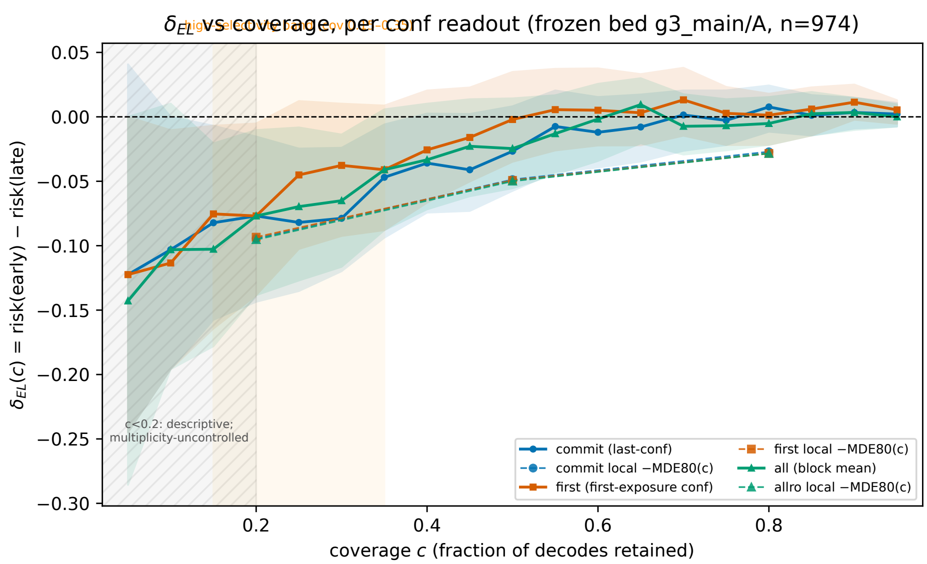


Figure 2: Early-minus-late risk $\delta_{\rm EL}(c)$ for all three readouts, with decode-clustered 95% CIs. Dashed, marker-matched curves show the coverage-local MDE80$(c)$ at registered anchors; no constant MDE band is projected across the axis. The hatched $c < 0.2$ region is descriptive and multiplicity-uncontrolled. **Takeaway:** the negative point pattern is concentrated at high selectivity, but the registered family has no survivor and the tail is not evidence of practical equivalence.

**Why not conformal or RCPS on this frozen $k = 1$ stream.** Table 11 positions each of these lines by the unit its confidence score is defined on, when the evidence is aggregated, the selection endpoint it feeds, and how aggregate uncertainty is treated; the junction measured here—which in-decoding aggregation of a masked diffusion decoder's confidence serves as the decode-level selective score, with paired clustered inference on both rank and risk—is not covered by any of them. Distribution-free risk-control frameworks (conformal prediction (Vovk et al., 2005), risk-controlling prediction sets (Bates et al., 2021; Angelopoulos & Bates, 2023), and selective-prediction risk control (Geifman & El-Yaniv, 2017)) deliver marginal guarantees on future draws of a *pipeline-defined* score. The object audited here is narrower: conditioning on a *frozen*, single-pass $k = 1$ decode stream, which of two in-decoding aggregations is the better selective score *inside this dataset*. No calibration split exists—each prompt contributes exactly one decode—and the identity of the "working" score is

Table 10: Frozen-bed registry and answer-extractor audit. **Takeaway:** the audit was executed on the frozen prediction text in this refine pass; branch frequencies, unparseable count, and the manual audit are now populated rather than asserted absent, but the generation and extraction provenance of the original bed remain incomplete.

| Field | Frozen setting or audit result |
|---|---|
| Sample | 974 problems; one decode/problem; 8 blocks; block length 32; NFE 64 |
| Readouts | `commit`=last; `first`=first; `all`=block mean |
| Judge | `#### N`>boxed>last-number; tolerance $10^{-6}$ |
| Base score | accuracy 0.7053; error 0.2947 |
| Fallback frequency | `hash` $2/974$ (0.21%); `last_num` $257/974$ (26.4%); `boxed` $714/974$ (73.3%); absent $1/974$ — rerun on the frozen text, this refine |
| Unparseable count | $8/974$ (0.82%; extract-returns-None) — rerun on the frozen text, this refine |
| Manual equivalence audit | $n = 200$ audit against a canonical-independent extractor; label-flip rate 30% (driven by `last_num`-branch disagreements); counts as a noise *upper bound*, not a calibrated estimate |
| Inference | Decode-cluster bootstrap; paired vector; fixed reported seeds |

Table 11: Closest-work positioning. Columns: the unit the confidence score is defined on, when the evidence is aggregated, what selection endpoint it feeds, and how aggregate uncertainty is treated. **Takeaway:** the audit of *which in-decoding aggregation of a masked diffusion decoder's confidence* to use as a decode-level selective score, with paired clustered inference on both rank and risk scales, is the specific junction not covered by prior work.

| Work | Score unit / timing | Selection endpoint | Uncertainty treatment |
|---|---|---|---|
| LLaDA (Nie et al., 2025) (the frozen bed) | token; per-step, in decoding | remasking order | none at decode level |
| Fast-dLLM (Wu et al., 2025) | token (action); per-step, in decoding | parallel unmask count | none at decode level |
| Early halting (Lo Cicero Vaina et al., 2023) | sequence / step; in decoding | stop / continue | heuristic threshold |
| Self-consistency (Wang et al., 2023) | candidate answer; post-hoc over $k$ samples | majority vote | vote frequency |
| Calibration (Guo et al., 2017) | single-pass prob; post-hoc | confidence reliability | temperature / ECE |
| Selective classification (Chow, 1970; Geifman & El-Yaniv, 2019) | single-pass feature/prob; post-hoc | abstain vs predict | risk–coverage curve |
| Failure prediction (Hendrycks & Gimpel, 2017; Kadavath et al., 2022) | response; post-hoc | abstain vs accept | AUROC / calibration |
| **This work** | **in-decoding revision; first / last / mean exposure** | **risk-gated decode selection** | **paired clustered bootstrap, dual scale** |

itself the unknown. Splitting the stream post-hoc to fit a conformal threshold would (i) break the paired structure that underlies every claim in this paper, and (ii) only certify the calibration subset at the coverages where it already has low error, collapsing to the descriptive bounds of Table 4. We therefore audit the readout with paired clustered resampling on the full stream, and leave guarantee-based control to the preregistered replication bed (§6), where a held-out split is part of the design.

## 6 LIMITATIONS

This is one model–task–temperature bed with no generation replication, no second dataset, and no causal intervention on readouts. Block number and canvas position are not crossed, so the early/late analysis is not a timing mechanism test. R2 depends on an unprotected $\mathrm{cov}_u$ correction; a 61-point residual-covariance sensitivity sweep over $m \in [0.85, 1.15]$ has now been run on the frozen bed (zero GPU, this refine) and is disclosed below, but the claim remains an *analysis*-level statement, not a causal one. R5's fine grid is descriptive outside the registered anchors and has no family survivor. The H2 ratio is a strict, observed-SE robustness flag, not evidence of absence; only the reported TOST result supports equivalence. The unrecorded decoder configuration and the original extraction-stage audit prevent regeneration. A post-hoc rerun of the canonical extractor on the frozen prediction text (this refine) populated branch census, unparseable count, and an $n = 200$ manual audit (Table 10); constraining the $\pm 0.0278$ headline by paired-agreement noise beyond that audit would require a calibrated human gold set, which is not resource-feasible here.

## 7 Conclusion

This bed supports a narrow conclusion: `commit` and `all` separate in rank units, while their incremental risk utility is neither resolved at the lower coverages nor established absent. A decisive test requires a fully recorded replication on a second model and task with $n \approx 1960$–$2820$, repeated decode seeds, archived remasking policy and prompt, and the same paired `commit`-vs-`all` endpoint. R3 would be falsified if that preregistered replication showed a practically relevant risk increment whose TOST interval lies outside $\pm 1$ pp, or if rank separation disappeared under the aligned contrast. Recording the decoder and extractor is therefore part of the measurement, not optional reproducibility metadata.

**Data and analysis release.** The analysis layer—-runtime metadata, schemas, analysis scripts, a co-located manifest, and the fixed identifiers of the frozen event stream (SHA256 `6e39e0ad3d4a42340ab6ef31474a20a2bdf7356dac9c9dbf6c0de63968b85972`; metadata `2b5c1f9d116257a7ccab6403c619cf0b74526472e4cabe110ed8a97cdf6258a5`)—-is released without precondition, and the stream contains no proprietary decoder internals. Release of the raw event-stream bundle itself remains subject to the project data steward's decision, as stated in §4. The remasking policy, decode seed, and prompt configuration were *not* recorded at decode time, so the decoder cannot be re-executed; what is auditable is the frozen-stream *analysis* protocol, not the decoding itself. Consequently this release is a frozen-stream analysis protocol, not a claim of model-internal access and not a reproducible decode.

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## A READOUT AND INFERENCE ALGORITHMS

### A.1 EVENT STREAM AND READOUT CONSTRUCTION

Each decode produces an event stream of records $(\mathrm{pos}, \mathrm{step}, \mathrm{conf})$, one per revision of position $\mathrm{pos}$: an event at step $s$ records the model's confidence $\mathrm{conf} \in (0, 1]$ in the current token assignment of that position. A position is *first exposed* at its first recorded event and *committed* at its last; a position is *revised* if and only if $|E_{d,p}| > 1$. Positions with no recorded event carry no score and are excluded. The three readouts and the decode-level selective score are built by Figure 3. Event extraction from the frozen decoder dump is unchanged from the frozen asset; only the aggregation is restated here.

```
Algorithm 1  Readout construction and decode-level selective score
Input : event stream E = {(d, pos, s, v)} for decode d;
        readout r in {commit, first, all}; blocks B_1..B_8; coverage c.
Output: selective score S_d^(r); retained set at coverage c.
for each (d,pos) do                          # pos-level readout
   if r = commit:  c_(d,pos) <- v at max s in E_(d,pos)  (last rev.)
   if r = first:   c_(d,pos) <- v at min s in E_(d,pos)  (first exp.)
   if r = all:     c_(d,pos) <- mean of v in E_(d,pos)  (rev. mean)
for each (d, block b) do                     # block-level aggregation
   R_(d,b)^(r) <- mean of c_(d,pos) over pos in b with any event
for each d do                                # decode-level score
   S_d^(r) <- (1/8) * sum_b R_(d,b)^(r)
sort decodes by S_d^(r); ties by stable decode index (mergesort)
retain k = max(1, round(c * n)) top-ranked decodes
```

Figure 3: Reference pseudocode for the three confidence readouts and the decode-level selective score. An independent implementer given the public event stream and this rule alone reproduces the three readouts and rankings exactly.

**Repeated revisions, unexposed positions, and answer length.** Every grouped mean is defined over *positions with at least one event*; unexposed positions (padding or never-touched canvas) contribute no term. Revisions enter `all` as an unweighted arithmetic mean over that position's events, so a position revised twice and one revised five times contribute equally at the position level (each is one position); the real stream has between $1$ and $8$ events per exposed position (median $4.5$, Table-of-support in the release). Generated answers vary in length across decodes, so the number of exposed positions per block varies; because every normalization is a per-position arithmetic mean, the block readout is invariant to answer length. An exposed position always has a commit and a first value (its last and first event), so no readout is undefined.

### A.2 PAIRED AUROC AND CLUSTERED RISK INFERENCE

AUROC is computed on the *pooled* decode score $\mathrm{S}_d^{(r)}$ against the binary correctness label with mergesort tie handling. The aligned contrast in Table 5 is the paired difference $\mathrm{AUROC}(\texttt{commit})-$

AUROC(`all`) on identical decode-cluster bootstrap resamples; it is *not* a block-wise AUROC averaged across blocks (that is a different estimand, used only as a reported auxiliary). Risk@coverage uses the bootstrapped per-block risk defined by Equation 1 and averaged over the eight blocks. All confidence intervals are percentile bootstrap CIs over decode clusters (the independent unit is the decode), $B = 2000$ resamples, seed 20260825; first-/last-exposure tie order within a position is resolved by the event sort on $(\mathrm{pos}, \mathrm{step})$ and never affects a reported digit (retained-set tie audit in Table 4).

### A.3 Hand-verifiable worked example

One decode, one block of two exposed positions, events $(\mathrm{pos}, \mathrm{step}, \mathrm{conf})$: for position 1, $(1, 1, .6)$, $(1, 3, .8)$ (two revisions); for position 2, $(2, 2, .9)$ (never revised). First exposure $\rightarrow$ commit is read directly: position 1 is exposed at .6 and committed at .8; position 2 at .9 throughout. Then $\texttt{commit} = (.8 + .9)/2 = .85$, $\texttt{first} = (.6 + .9)/2 = .75$, $\texttt{all} = \big((.6 + .8)/2 + .9\big)/2 = .80$. With one block the decode scores equal these block values; ranking three such decodes by `commit` already separates them as .85 above a single-revision peer at .80 and a decaying one at .75. With one block the denominator 8 scales all readouts identically and leaves the ranking unchanged.

## B Inference details

Retained counts use banker's rounding with a floor at one; at $c = 0.2$, $\mathrm{round}(194.8) = 195$. Stable index order breaks exact score ties. The twin statistic is $p_{\mathrm{twin}} = 2 \min\{\widehat{\Pr}(\hat{\delta}_b > 0), \widehat{\Pr}(\hat{\delta}_b < 0)\}$ over paired bootstrap resamples. For label permutations we use the two-sided tail count on $|\Delta|$, $p = (1 + \sum_{b=1}^{B} \mathbf{1}\{|\Delta_b^{\pi}| \geq |\Delta_{\mathrm{obs}}|\})/(B + 1)$. The reported $B = 2000$ resolution is therefore expressed as $p \leq 1/(B + 1) \approx 5 \times 10^{-4}$, not as an exact $p = 0.0005$.

Table 12: Full R3 risk grid with conventional two-sided $p$ and H2 robustness flag. The H2 boundary is $p \approx 0.0051$; it is stricter than nominal detection. **Takeaway:** nominal detection and H2 robustness must be read separately.

| Anchor | $\hat{\Delta}$ | 95% CI | MDE80 | Ratio | $p_{2s}$ | Reading |
|---|---|---|---|---|---|---|
| intra $c = .2$ | $-.0058$ | $[-.0199, .0058]$ | .0161 | 2.79 | .3160 | CI $\ni$ 0; H2 PL |
| intra $c = .5$ | $-.0067$ | $[-.0131, .0008]$ | .0108 | 1.62 | .0835 | CI $\ni$ 0; H2 PL |
| intra $c = .8$ | $-.0008$ | $[-.0056, .0022]$ | .0029 | 3.60 | .4362 | CI $\ni$ 0; H2 PL |
| cross $c = .2$ | $-.0410$ | $[-.077, -.005]$ | .0513 | 1.25 | .0250 | detected; H2 PL |
| A0 full $c = .2$ | $+.2126$ | $[.125, .218]$ | .0660 | .31 | $< 10^{-18}$ | detected; H2 RES |
| A0 full $c = .5$ | $+.1407$ | $[.079, .147]$ | .0484 | .34 | $< 10^{-15}$ | detected; H2 RES |
| A0 full $c = .8$ | $+.0649$ | $[.038, .098]$ | .0431 | .67 | $2.5 \times 10^{-5}$ | detected; H2 RES |
| A1 shuffle $c = .2$ | $+.0038$ | $[-.064, .069]$ | .0953 | 25.39 | .9121 | CI $\ni$ 0 |
| A1 shuffle $c = .5$ | $+.0073$ | $[-.036, .048]$ | .0602 | 8.27 | .7348 | CI $\ni$ 0 |
| A1 shuffle $c = .8$ | $+.0079$ | $[-.025, .039]$ | .0459 | 5.79 | .6282 | CI $\ni$ 0 |

Table 13: Early/late anchors with two-sided normal-reference $p$. H2 `PL`/`RES` changes at $p \approx 0.0051$; conventional CI and Holm decisions remain separate. **Takeaway:** every registered early/late anchor remains H2 power-limited.

| Readout/$c$ | $\hat{\delta}$ | 95% CI | MDE80 | Ratio | $p_{2s}$ | H2 |
|---|---|---|---|---|---|---|
| commit/.2 | $-.0769$ | $[-.144, -.010]$ | .0944 | 1.227 | .0224 | PL |
| commit/.5 | $-.0267$ | $[-.058, .008]$ | .0487 | 1.824 | .1245 | PL |
| commit/.8 | $+.0077$ | $[-.012, .026]$ | .0271 | 3.518 | .4258 | PL |
| first/.2 | $-.0769$ | $[-.133, -.005]$ | .0934 | 1.215 | .0211 | PL |
| first/.5 | $-.0021$ | $[-.035, .035]$ | .0493 | 24.036 | .9072 | PL |
| first/.8 | $+.0013$ | $[-.021, .019]$ | .0284 | 22.136 | .8993 | PL |
| all/.2 | $-.0769$ | $[-.138, -.005]$ | .0953 | 1.239 | .0237 | PL |
| all/.5 | $-.0246$ | $[-.055, .014]$ | .0499 | 2.025 | .1665 | PL |
| all/.8 | $-.0051$ | $[-.022, .018]$ | .0284 | 5.539 | .6130 | PL |

Table 14 isolates the raw knife-edge family member from the registered family-level verdict.

Table 14: Knife-edge Monte Carlo audit for the raw $c = 0.2$ early/late family member. The registered family outcome remains empty; this table diagnoses resampling sensitivity rather than promoting a survivor. **Takeaway:** the smallest raw $p$ straddles the Holm boundary, but the registered family verdict stays empty.

| Audit | Frozen value | Reference | Reading |
|---|---|---|---|
| High-$B$ bootstrap | $B = 20{,}000$ | 10 seeds | CI excludes zero in all runs |
| Smallest-$p$ range | 0.0144–0.0174 | $\alpha/3 = 0.01667$ | Threshold-straddling |
| Mean smallest $p$ | 0.01636 | MC-SE 0.0011 | Not a stable survivor |
| CI endpoint drift | $\approx 3 \times 10^{-17}$ | discrete grid | Endpoints stable |
| Registered outcome | 0 survivors | Holm(3) | Retain boundary verdict |

## C AUROC-TO-RISK PROJECTION ASSUMPTIONS

The projection $\widehat{\Delta\text{risk}} = \Delta A(1 - c)d$ assumes monotone score ordering, a first-order Taylor approximation, independent decode clusters, and no calibration equality. It is neither a confidence bound nor a measured risk increment; Equation 1 remains the selection estimand.

**Derivation and references.** Write the risk at coverage $c$ as the conditional error over the retained set $\mathcal{S}_c = \{i : s_i \geq \tau_c\}$, with $\tau_c$ the per-decode score threshold that keeps a fraction $c$ of the mass. Differentiating under the dependence assumptions above (cf. Geifman & El-Yaniv, 2017; El-Yaniv & Wiener, 2010, for the risk-coverage family) plus the AUROC-to-tail decomposition of ranking losses (Agarwal et al., 2005; Clémençon et al., 2008) gives, to first order in a small score shift $\Delta A$,

$$\widehat{\Delta\text{risk}}(c) \;\approx\; \Delta A \cdot (1 - c) \cdot d,$$

where $d = \Pr(i \in \mathcal{S}_c \mid y_i = 1) \cdot \big(1 - \Pr(i \in \mathcal{S}_c \mid y_i = 1)\big)$ is the dispersion term pushed into the frozen constant. The $(1 - c)$ factor is the risk-density slope of a threshold policy at coverage $c$; it doubles when coverage halves. We report $d$ only as a frozen-bed diagnostic and never as a new score.

## D R2 RESIDUAL-COVARIANCE SENSITIVITY

The sweep scales the residual-covariance term by a multiplier $m$: $\widehat{\text{cov}}_u \mapsto m\,\widehat{\text{cov}}_u$. At the reported correction the upper bound is 0.979; multiplying by 0.9038 returns the point calculation to $\rho = 1$, while the CI-upper calculation crosses near 0.935. In this refine pass the $[0.85, 1.15]$ sweep was executed on the frozen events stream (61 points, step 0.005; each point a paired $B = 2000$ decode-cluster bootstrap; seed 20260823; asset `r211_r2_covu_sweep_v6`). Representative rows are in Table 15. The point estimate crosses $\rho = 1$ at $m \approx 0.905$; the 95% CI upper excludes $\rho = 1$ for $m \gtrsim 0.94$; and $\Pr(\rho < 1) \geq 0.99$ for $m \gtrsim 0.945$. The sweep therefore reproduces the breakeven diagnostic and the claim narrowing recorded in the ledger, without supplying a ground-truth $\rho$ for the latent structure.

Table 15: R2 residual-covariance sensitivity sweep on the frozen events stream (61 points; paired decode-cluster bootstrap, $B = 2000$, seed 20260823). $\rho$ denotes the disattenuated correlation and $m$ the multiplier on $\widehat{\text{cov}}_u$. **Takeaway:** the point estimate crosses $\rho = 1$ near $m \approx 0.905$, the CI upper excludes $\rho = 1$ from $m \approx 0.94$, and $\Pr(\rho < 1) \geq 0.99$ from $m \approx 0.945$; the result is an analysis-level sensitivity statement over the frozen asset, not a causal one.

| $m$ | $\widehat{\rho}(m)$ | 95% CI | $\Pr(\rho < 1)$ | Note |
|---|---|---|---|---|
| 0.85 | 1.018 | $[1.007, 1.029]$ | 0.00 | CI strictly above 1 |
| 0.905 | 1.000 | $[0.988, 1.010]$ | 0.52 | point-$\rho$ crosses 1 |
| 0.94 | 0.988 | $[0.977, 0.999]$ | 0.99 | CI upper returns to 1 |
| 0.945 | 0.986 | $[0.975, 0.997]$ | 0.99 | $\Pr(\rho < 1) \geq 0.99$ |
| 1.0 | 0.968 | $[0.956, 0.979]$ | 1.00 | nominal (assumed correction) |
| 1.15 | 0.919 | $[0.903, 0.934]$ | 1.00 | far end of requested range |

## E REGISTRATION AND PROVENANCE

Table 16 binds each claim to its frozen evidence path and timing.

Table 16: R1–R5 registration ledger. Paths are relative to the run root. "Post" means fixed after the first computation and is not presented as preregistration. **Takeaway:** evidence paths and timing are claim-specific; post-alignment analyses are not preregistrations.

| ID | Hypothesis and verdict | Evidence path | Registration/time relation |
|---|---|---|---|
| R1 | $\delta_{EL}(.2) < 0$; early–late; decode; zero. *Verdict:* lattice equality; sets differ | `agents/A23/workspace/results/r19_crossreadout_family.json` | r16/r17 before r19; grid/family fixed |
| R2 | corrected $\rho < 1$; commit–all; decode; unity. *Verdict:* not robustly separated | `agents/A23/workspace/results/r15_singlefactor_pointpred.json` | r15 model fixed before result; sensitivity post |
| R3 | $\Delta^{intra} \neq 0$; commit–all; decode; zero. *Verdict:* rank RES; risk unresolved | `agents/A23/workspace/results/r204_p1_paired_commitall/r204_p1_paired_commitall.json` | scope/grid before r204; paired AUROC post-alignment |
| R4 | first preserves early/late sign; decode; zero. *Verdict:* only .2 lattice-identical | `agents/A23/workspace/results/r17_window_desaturation.json` | r17 before first computation |
| R5 | effect localized to high selection; decode; zero. *Verdict:* no family survivor | `agents/A23/workspace/results/r21_coverage_shape_scan.json` | fine grid descriptive; Holm family pre-fixed |

*Run-ID map.* Each frozen asset links to the recorded output it contributed; IDs are omitted from the main verdict matrix.

| ID | Output | ID | Output |
|---|---|---|---|
| r16 | early/late decomposition | r17 | first-readout desaturation |
| r19 | three-readout family | r21 | descriptive coverage curve |
| r56 | full-width MDE correction | r106 | complete MDE/control grid |
| r197 | label-permutation Type-I | r198 | conservative calibration reading |
| r202 | twin-$p$ and tie audit | r204 | aligned paired contrasts |
| r205 | joint/calibrated/Jaccard/R5 | g3_main/A | frozen event-stream arm |
| partD | registered within-block risk | P1 sensitivity | $\mathrm{cov}_u$ breakeven |

Scope and $\{0.2, 0.5, 0.8\}$ were fixed before the registered R3 risk computation; the Holm family and reported seeds were fixed in the corresponding run files. The practical bound $\delta^* = 1$ pp was fixed before the TOST recomputation but after the original effect estimates. The 1.2 pp AUROC projection is post-hoc and explicitly descriptive. The aligned paired AUROC, calibrated central band, joint endpoint, and Jaccard audits were introduced during revision on the same frozen stream; they are post-hoc analyses, not prospective registrations.

## F JUDGMENT-HYGIENE CHECKLIST

H1 separates rank from utility; H2 treats full-width MDE as sensitivity, not post-hoc power; H3 desaturates; H4 reports intervals; H5 crosses readout and coverage; and H6 audits ties, retained sets, calibration, and reproduction. Their assembly is the reusable protocol.